A Geometric Analysis of Linear Brain Dynamics: Quantifying the Interplay Between Conservative and Dissipative Forces in State-Space Models of Resting-State fMRI

Supervisor: Prof. Eleonora Maggioni
Author: Maurizio Tirabassi
Co-Supervisor: Dr. Danilo Benozzo

The brain is a living system.

The brain consumes energy to perform biological functions.

At the microscale, energy is a well-defined quantity [1].

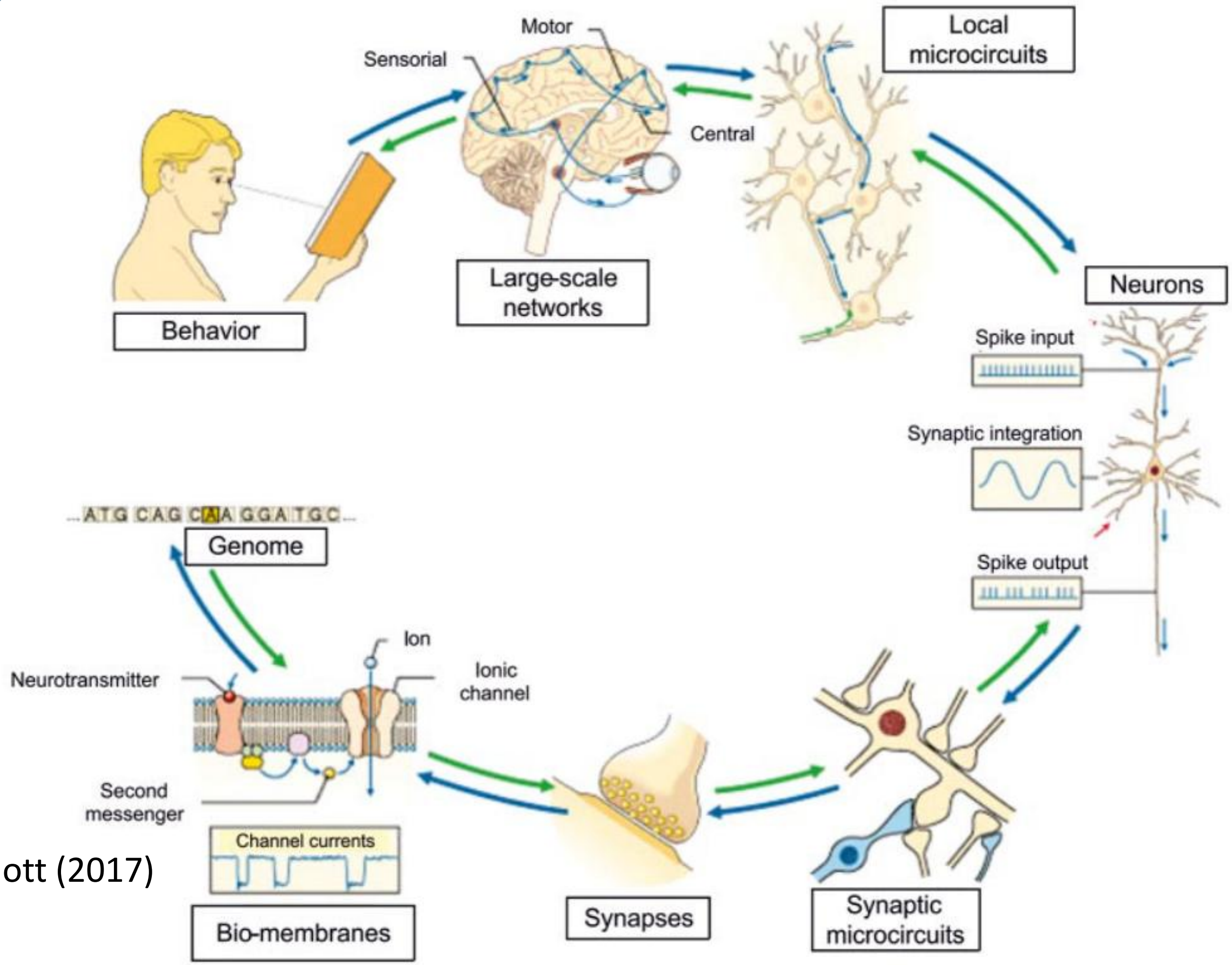


Figure: D'Angelo & Wheeler-Kingshott (2017) *La Rivista Del Nuovo Cimento*

Macroscale time-irreversibility is linked to microscale energy consumption.

The time-irreversibility of macroscale physiological recordings establishes a lower bound to the measure of microscale energy consumption [1].

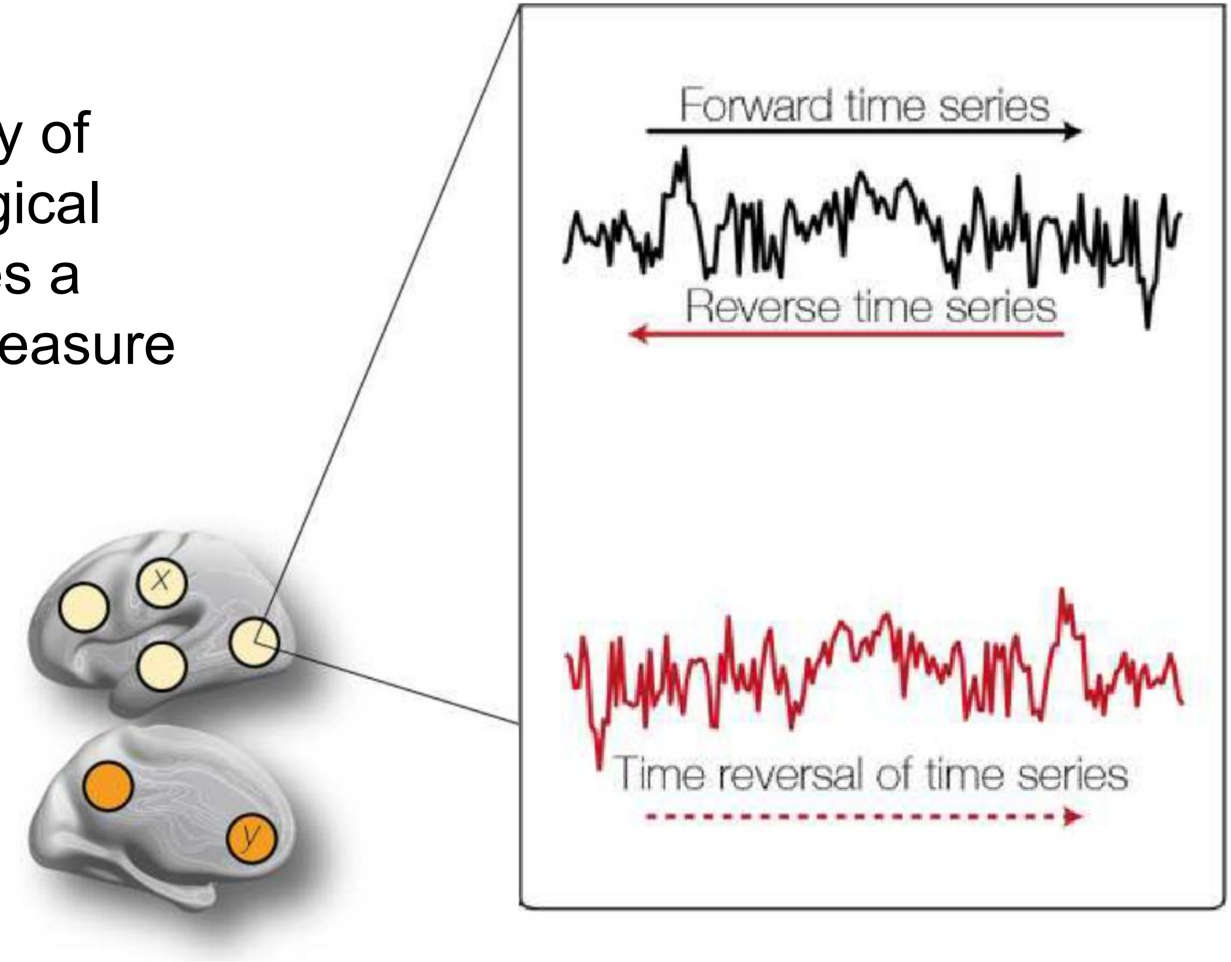


Figure: Kringelbach et al. (2023) *Science Advances*

Time-irreversibility correlates with states of consciousness and pathology [2].

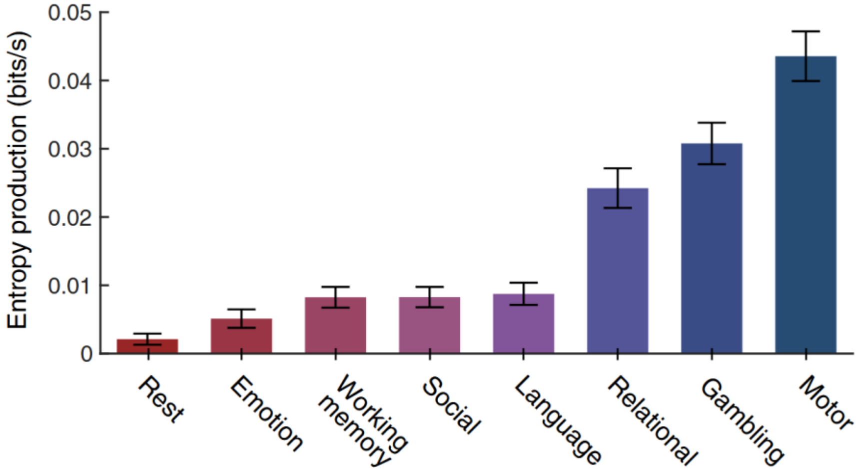


Figure: Lynn et al. (2021) *PNAS*

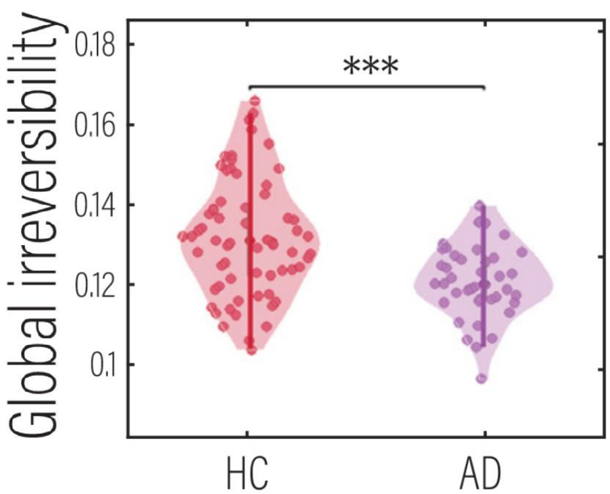


Figure: Cruzat et al. (2023) *The Journal of Neuroscience*

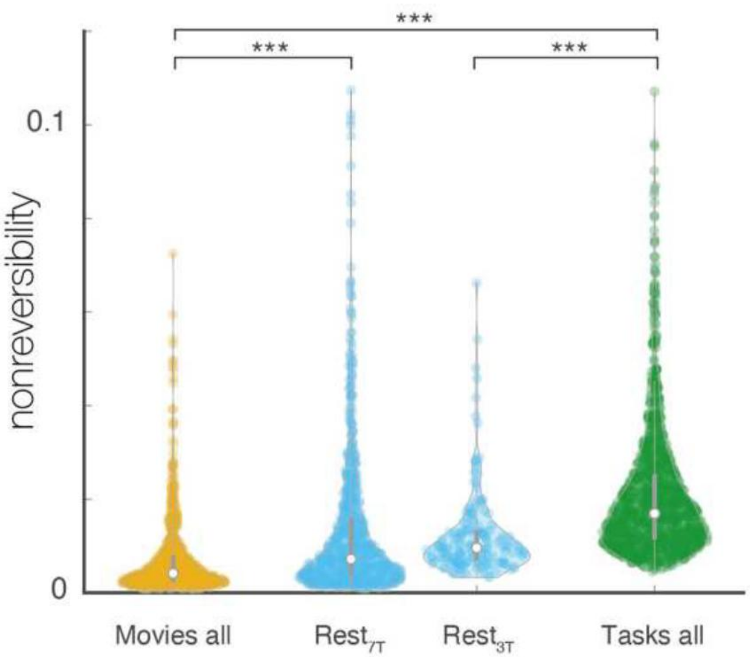
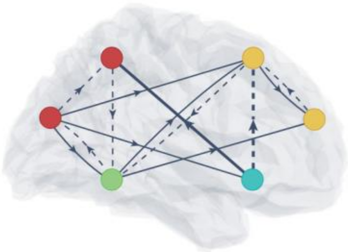


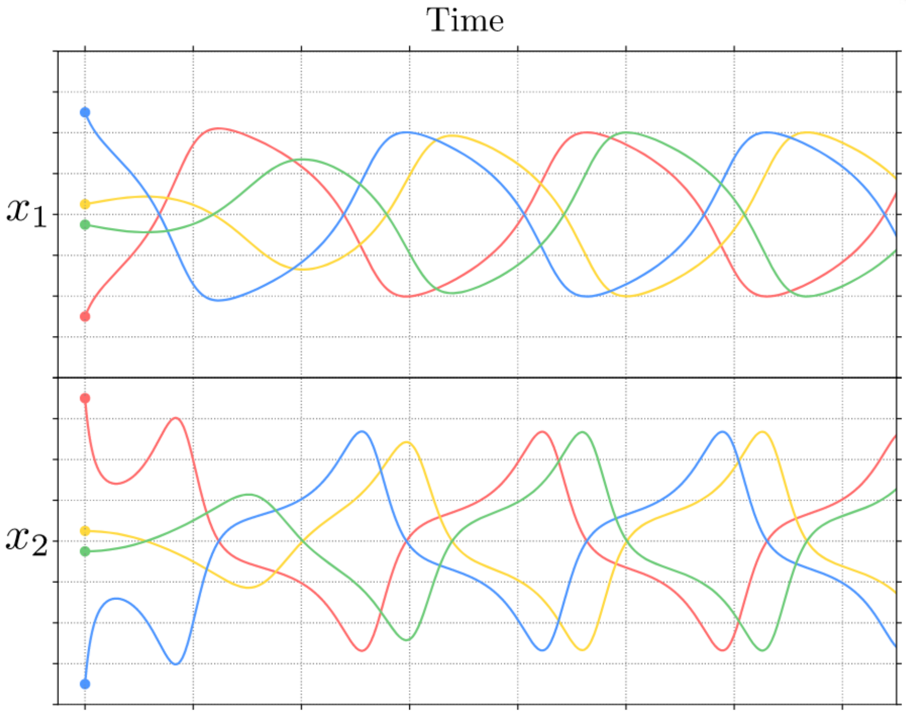
Figure: Kringelbach et al. (2023) *Science Advances*

Dynamical systems are used to model neural activity.

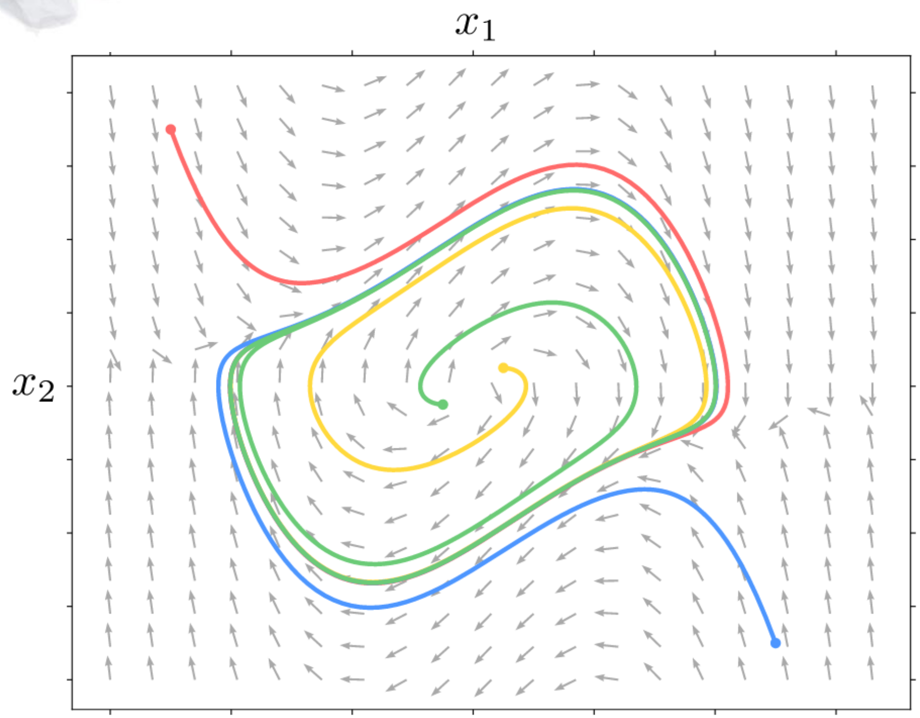
Figure: Medrano et al. (2023) *NETN*



$$\dot{x}(t) = f\big(x(t)\big)$$



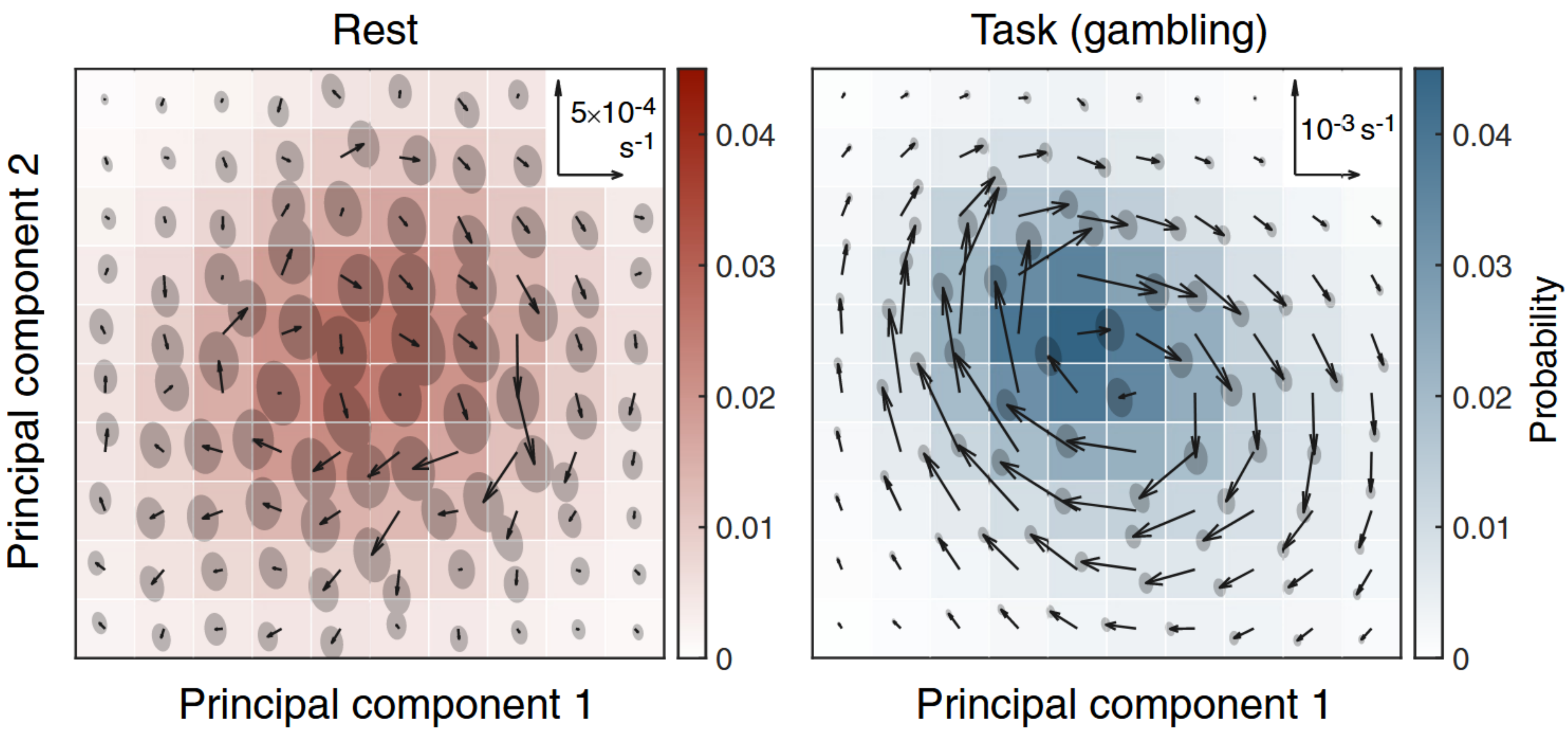
Trajectories in the time domain



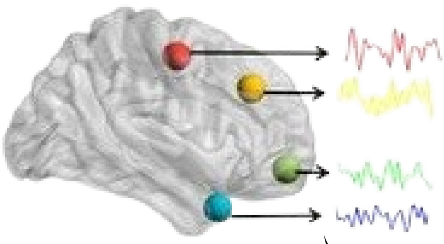
Flow and trajectories in state space

Time-irreversibility manifests as rotation in state space [1].

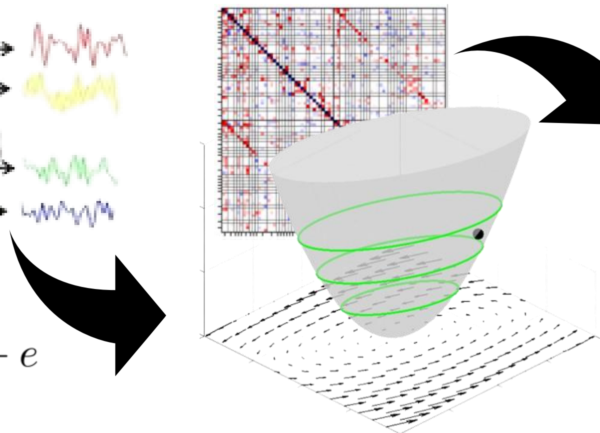
Figure: Lynn, C. W. et al. (2021) *PNAS*

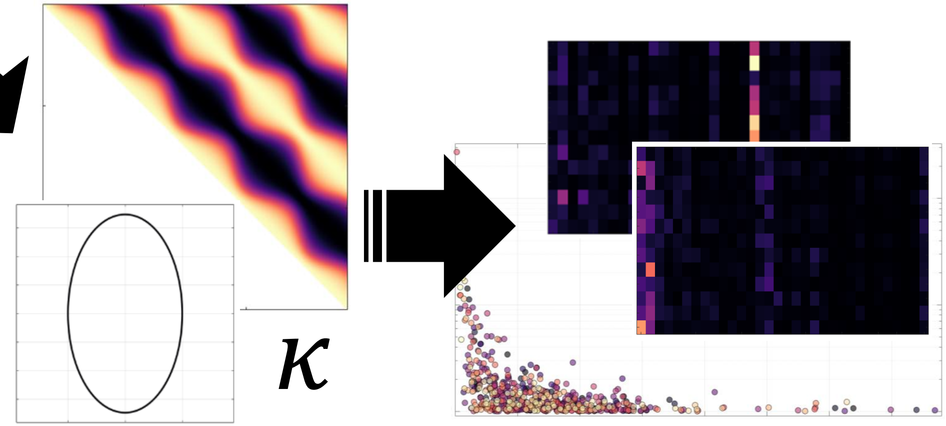


This thesis explores the gap between observed dynamics and underlying mechanics.



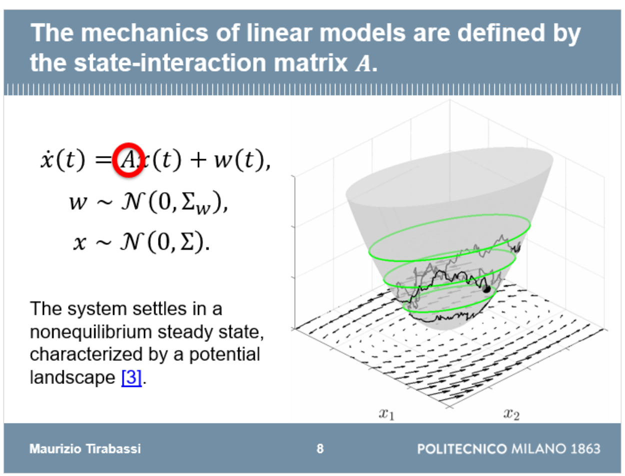
$$\dot{x} = Ax + w$$
$$y = (x * h) + e$$





Theoretical Framework

Application in Neuroscience

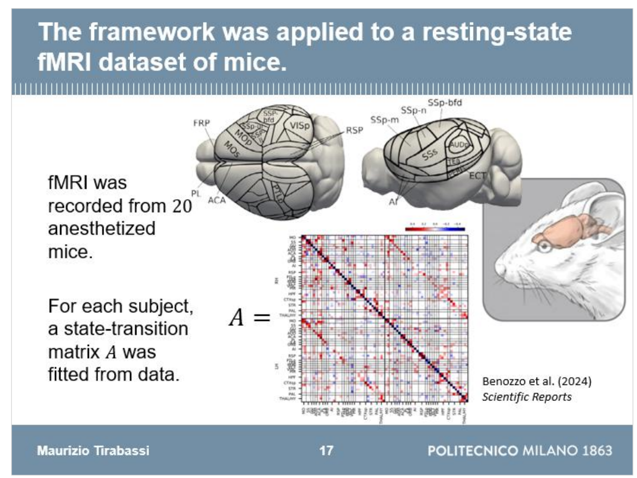


The mechanics of linear models are defined by the state-interaction matrix A.

$$\dot{x}(t) = Ax(t) + w(t),$$
$$w \sim \mathcal{N}(0, \Sigma_w),$$
$$x \sim \mathcal{N}(0, \Sigma).$$

The system settles in a nonequilibrium steady state, characterized by a potential landscape [3].

Maurizio Tirabassi 8 POLITECNICO MILANO 1863



The framework was applied to a resting-state fMRI dataset of mice.

fMRI was recorded from 20 anesthetized mice.

For each subject, a state-transition matrix A was fitted from data.

$A =$

Benozzo et al. (2024) *Scientific Reports*

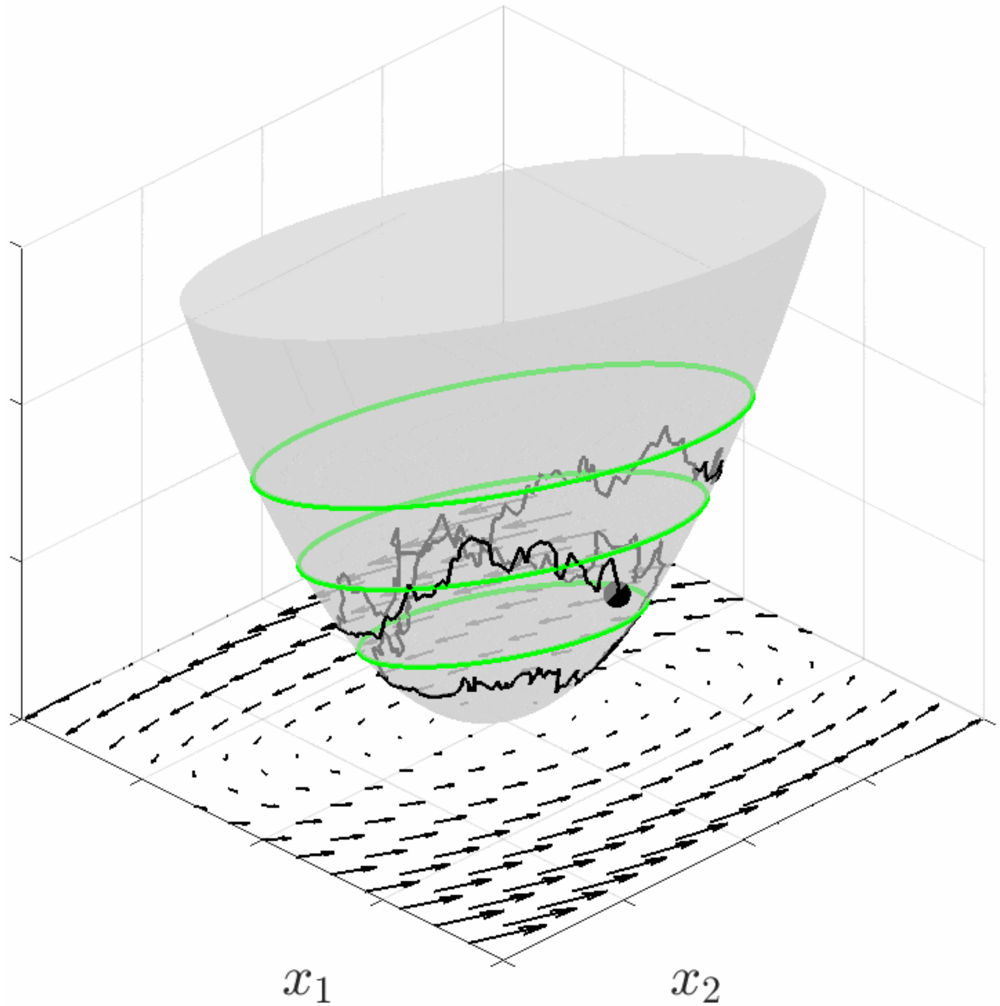
Maurizio Tirabassi 17 POLITECNICO MILANO 1863

The mechanics of linear models are defined by the state-interaction matrix A.

$$\dot{x}(t) = A x(t) + w(t),$$

$$w \sim \mathcal{N}(0, \Sigma_w),$$

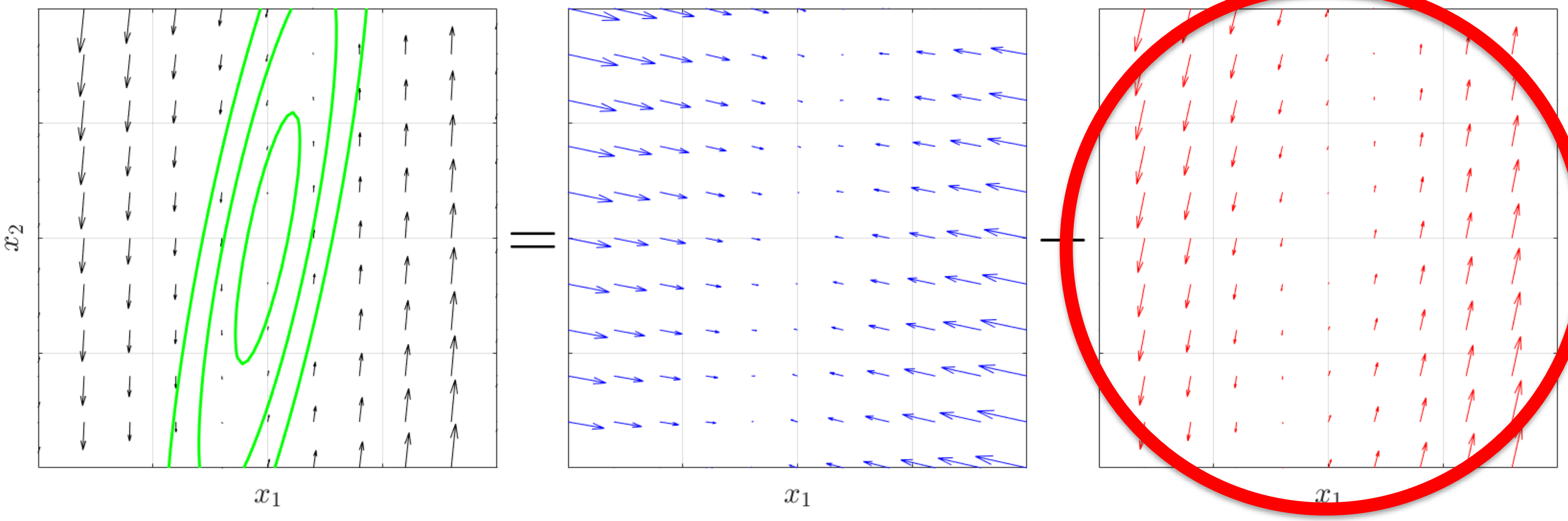
$$x \sim \mathcal{N}(0, \Sigma).$$



The system settles in a nonequilibrium steady state, characterized by a potential landscape [3].

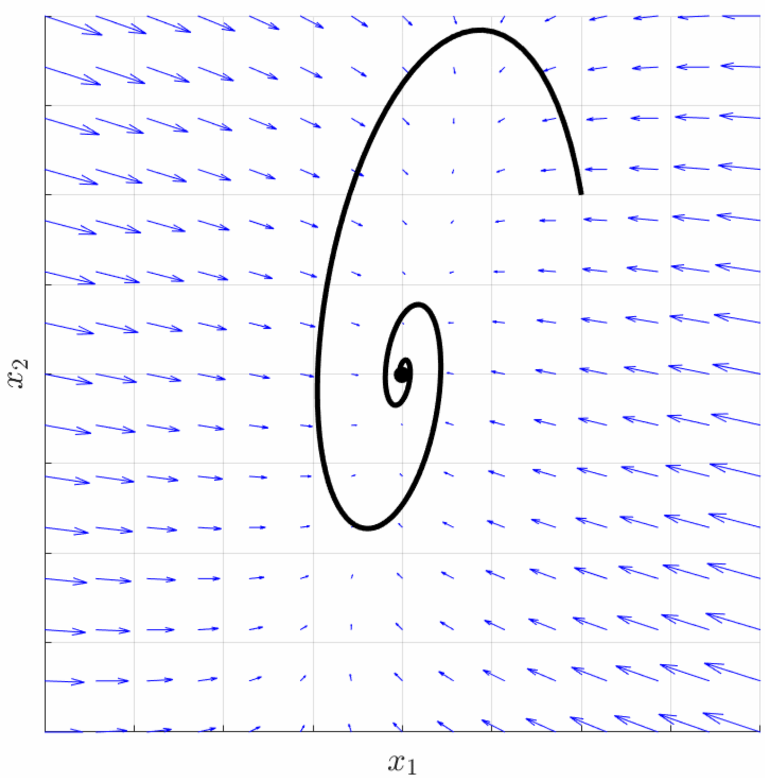
A is composed of dissipative and conservative fields.

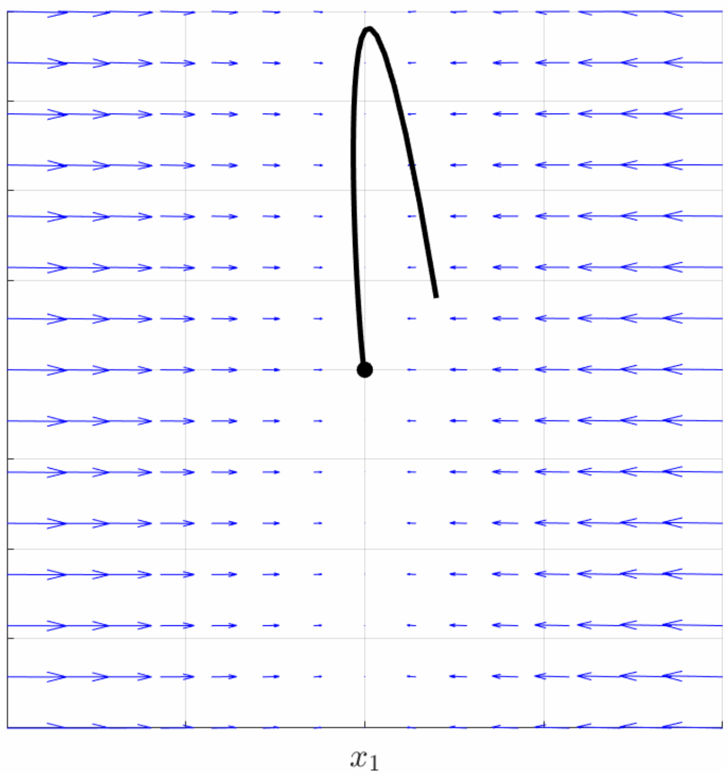
$$A = -\frac{1}{2}\Sigma_w \Sigma^{-1} + S\Sigma^{-1}$$



The skew-symmetric matrix S encodes time-irreversibility and drives the conservative flow [3].

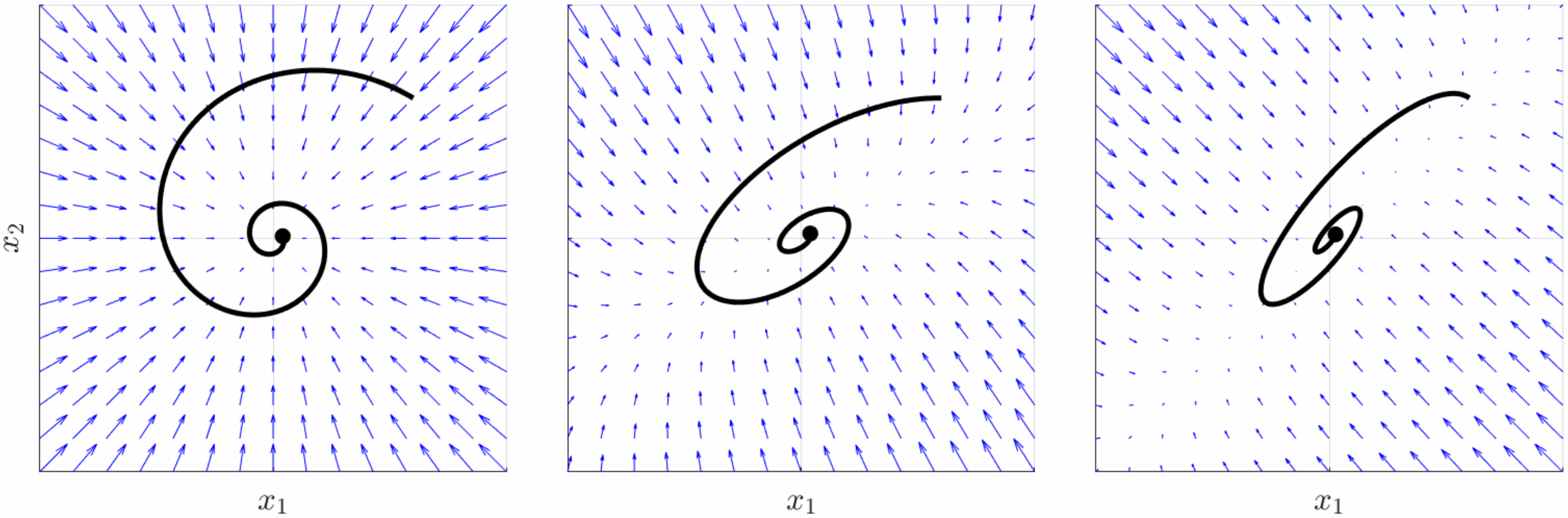
$S \neq 0$ does not guarantee oscillatory dynamics.





Nonzero S is a necessary but not sufficient condition for the onset of oscillatory dynamics (complex conjugate eigenvalues of A).

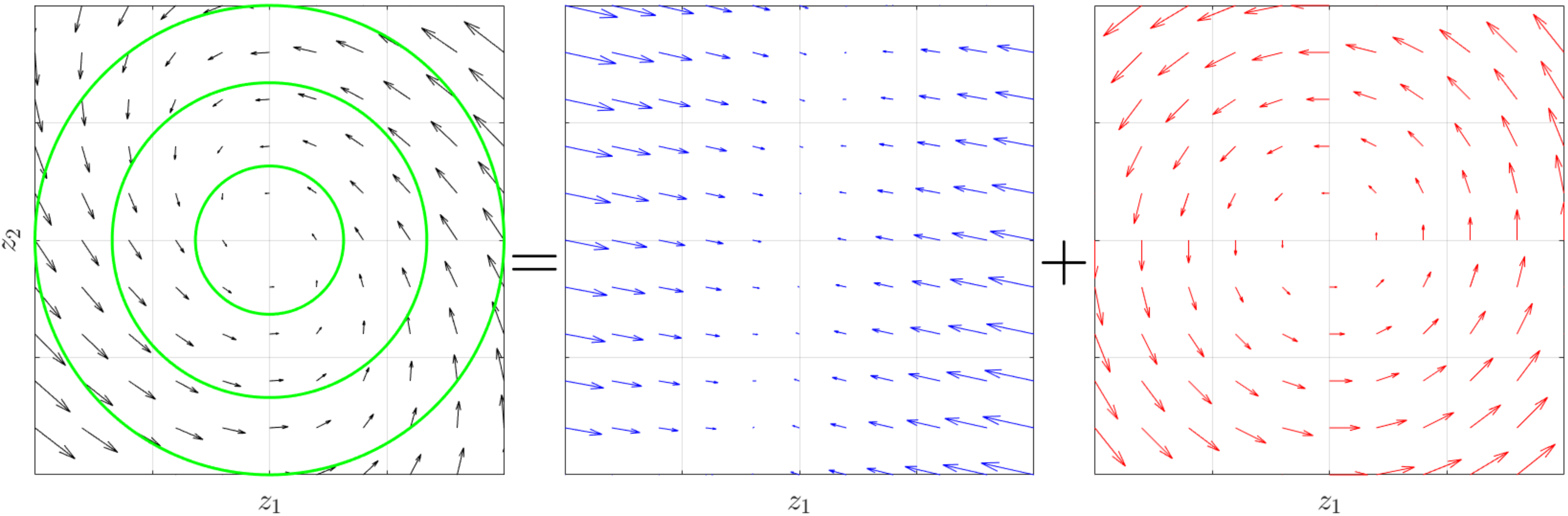
$S \neq 0$ can yield different oscillatory dynamics.



A specific frequency (ν) does not map to a unique mechanism.

Non-commutativity describes the interplay between dissipative and conservative forces.

$$z(t) = \Sigma^{-1/2}x(t), \quad \tilde{A} = \underbrace{-\frac{1}{2}\Sigma_w\Sigma^{-1}}_{\tilde{D}} + \underbrace{\Sigma^{-1/2}S\Sigma^{-1/2}}_{\tilde{S}}$$

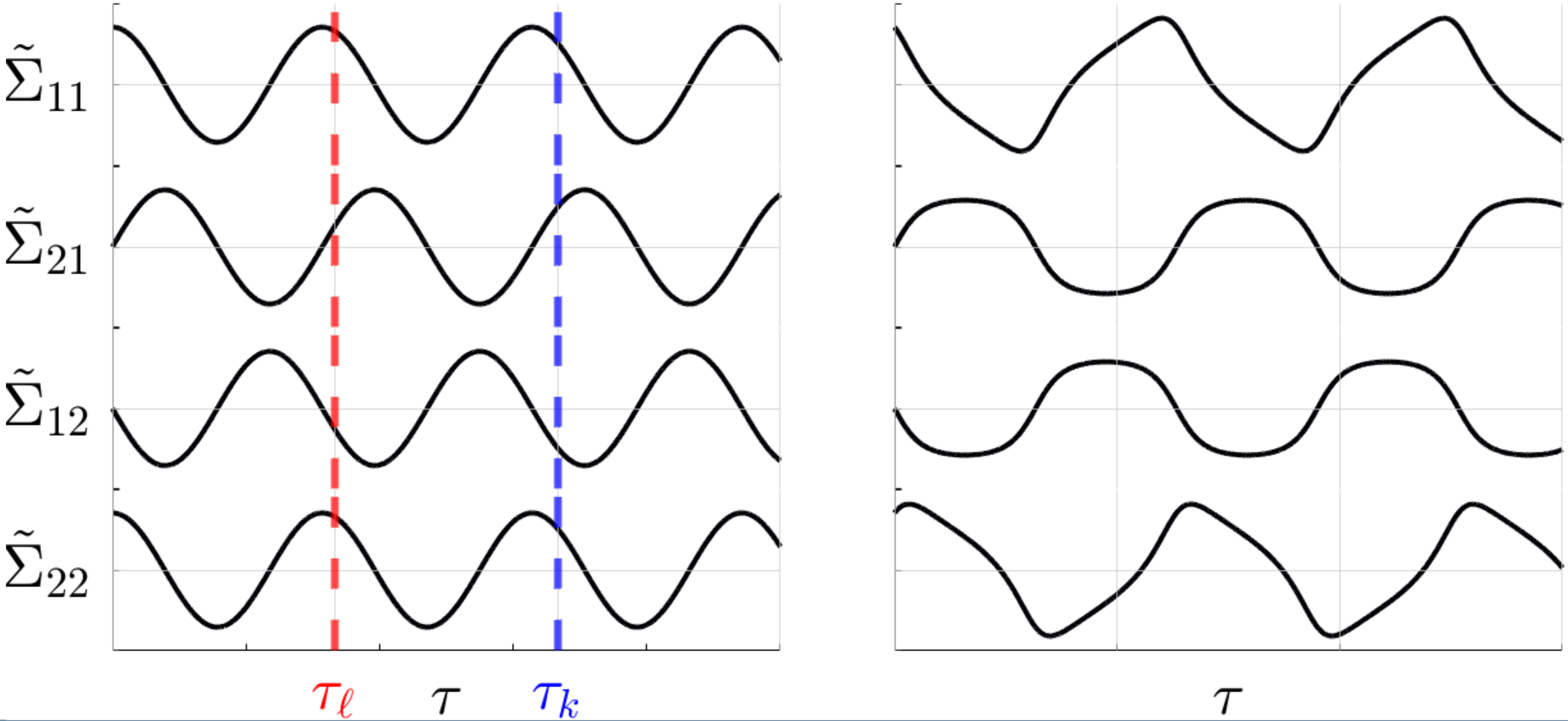


$$\tilde{S}\tilde{D} \neq \tilde{D}\tilde{S}$$

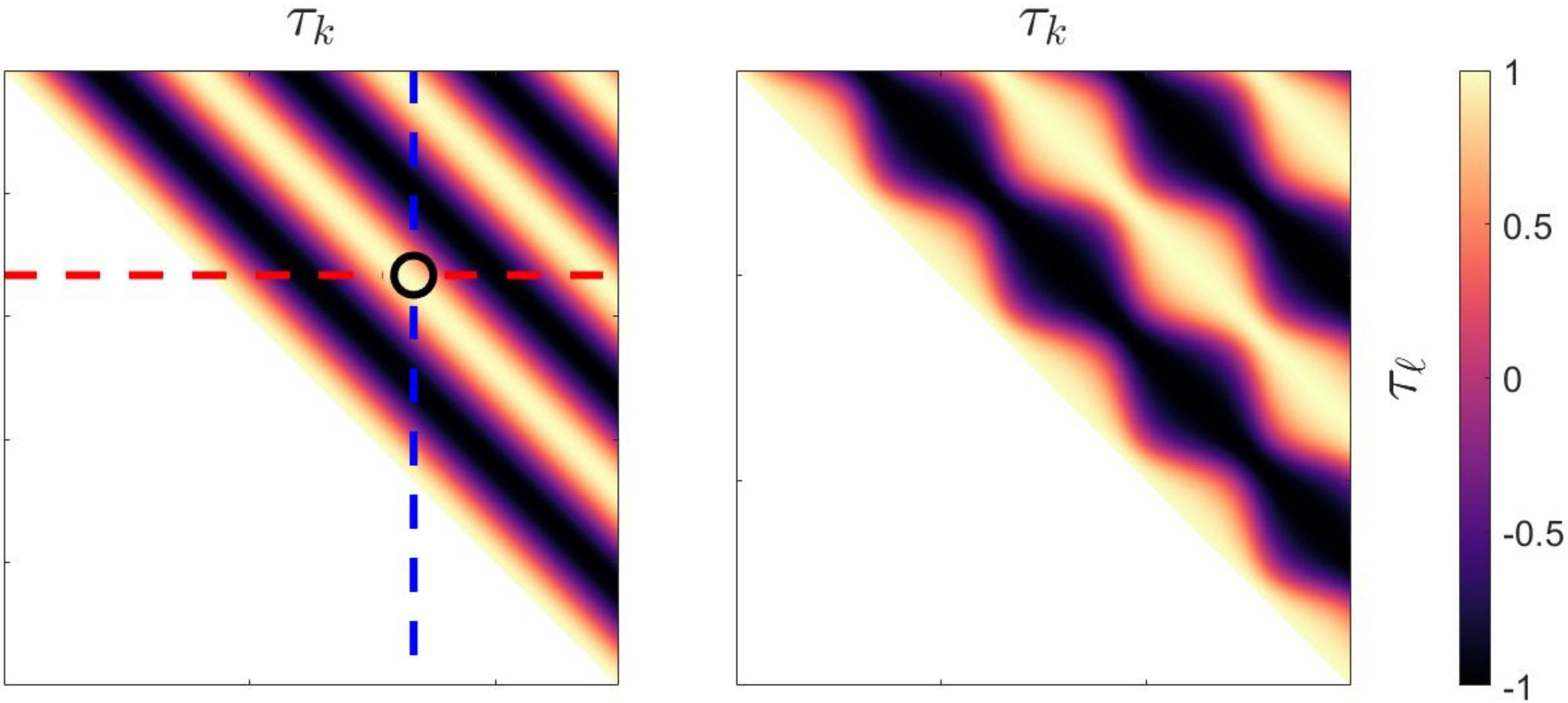
The time-lagged covariance matrix encodes the dynamics.

$$\tilde{\Sigma}(\tau) = e^{\tilde{A}\tau}$$

$$\tilde{S}\widetilde{D} = \widetilde{D}\tilde{S} \qquad \qquad \tilde{S}\widetilde{D} \neq \widetilde{D}\tilde{S}$$



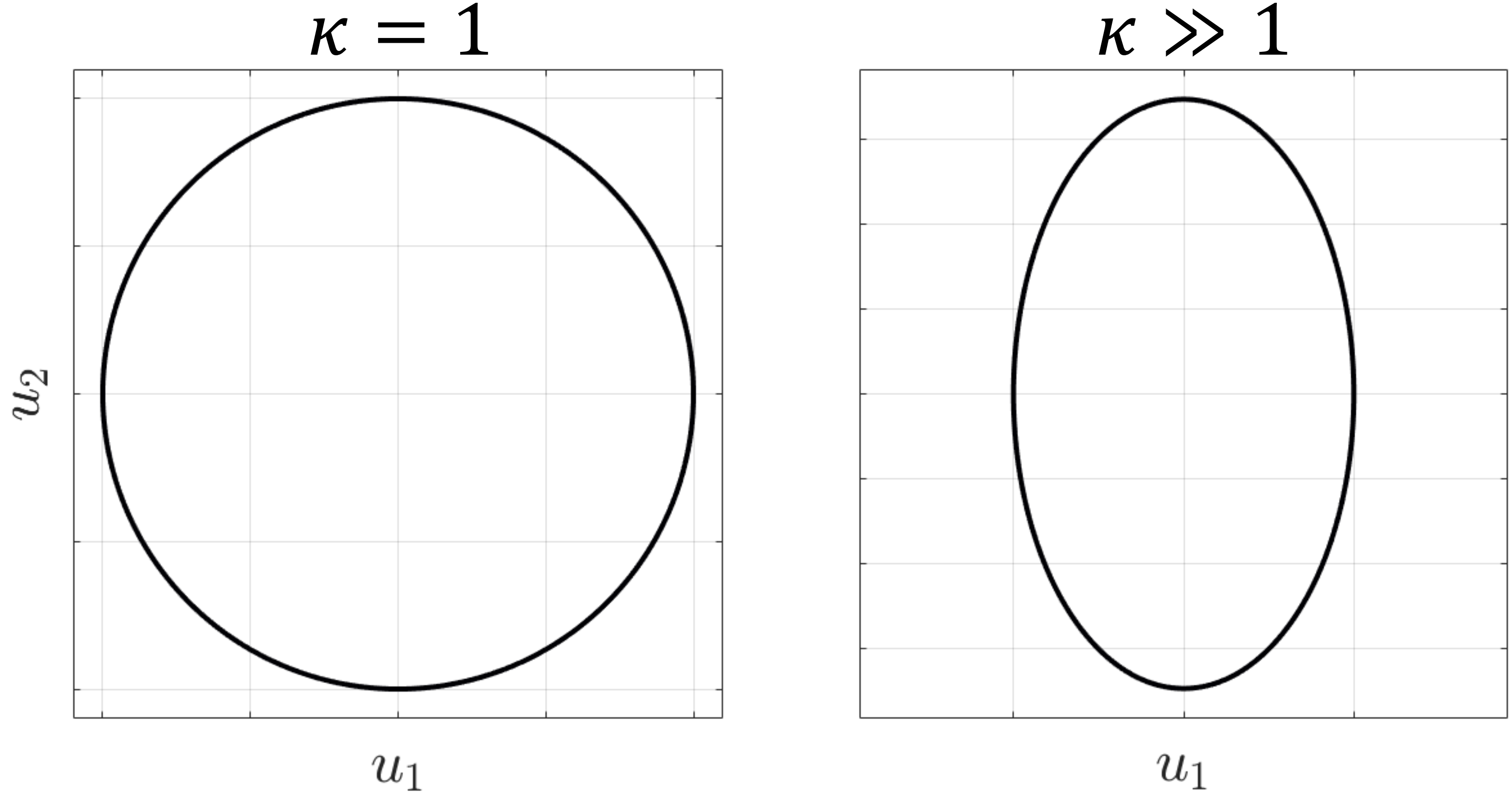
The Cross-Lag Similarity (CLS) quantifies non-commutativity.



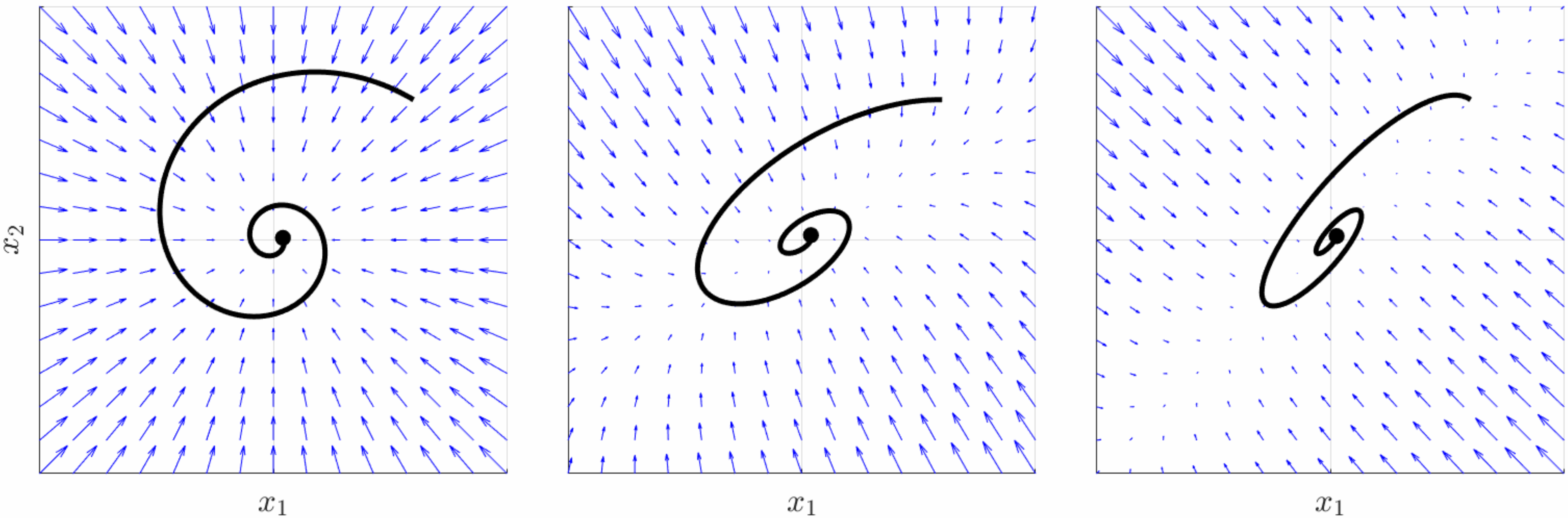
$$\cos_{k\ell}\theta \propto \cos(\nu\tau_k)\cos(\nu\tau_\ell) + \kappa \sin(\nu\tau_k)\sin(\nu\tau_\ell)$$

"Ellipticity" describes an oscillatory mode.

$$\cos_{k\ell}\theta \propto u(\tau_k)^\top u(\tau_\ell), \quad u(\tau) = [\cos(\nu\tau), \sqrt{\kappa}\,\sin(\nu\tau)]$$

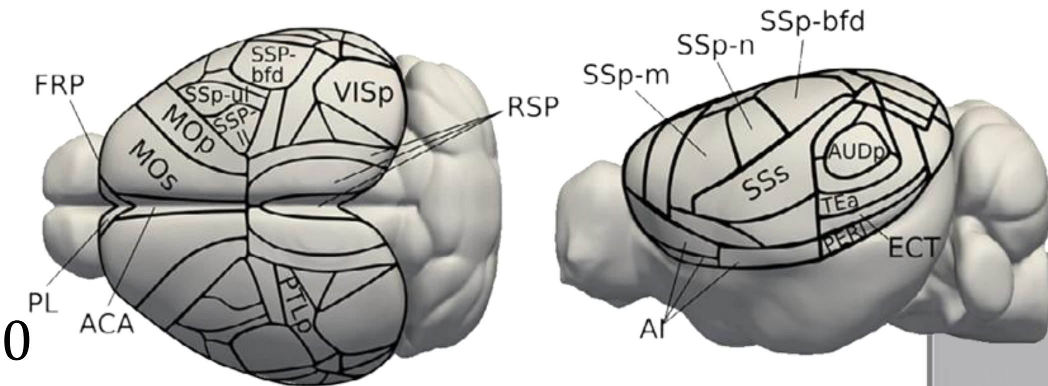


“Ellipticity” describes an oscillatory mode.



$\kappa = 1$ $\kappa = 1.04$ $\kappa = 1.08$

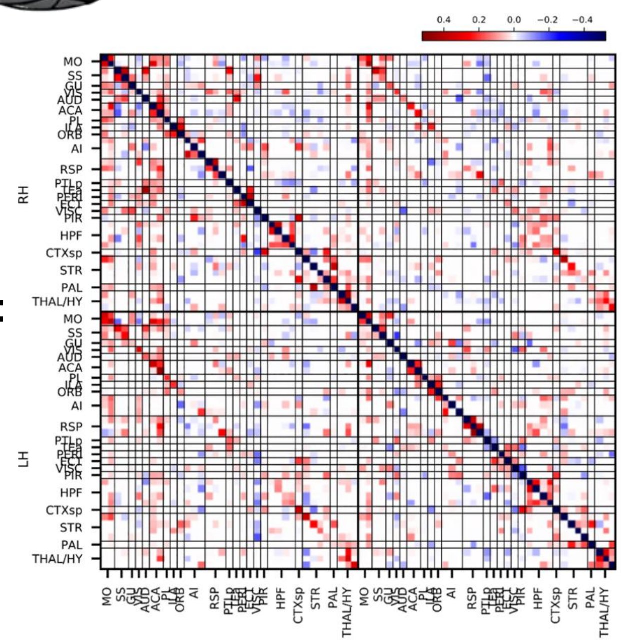
The framework was applied to a resting-state fMRI dataset of mice.



fMRI was recorded from 20 anesthetized mice.

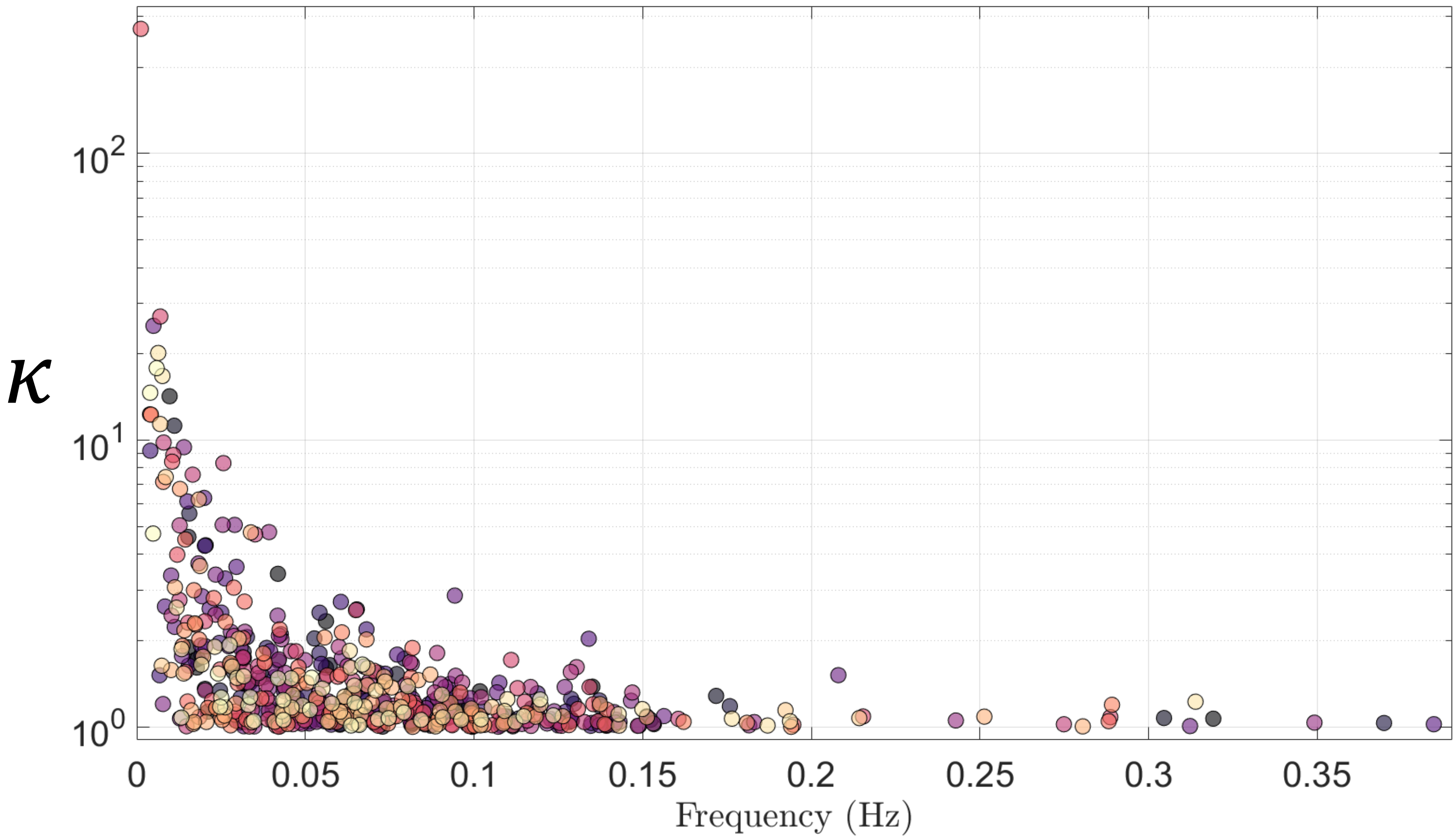
For each subject, a state-transition matrix A was fitted from data.

$$A =$$



Benozzo et al. (2024) *Scientific Reports*

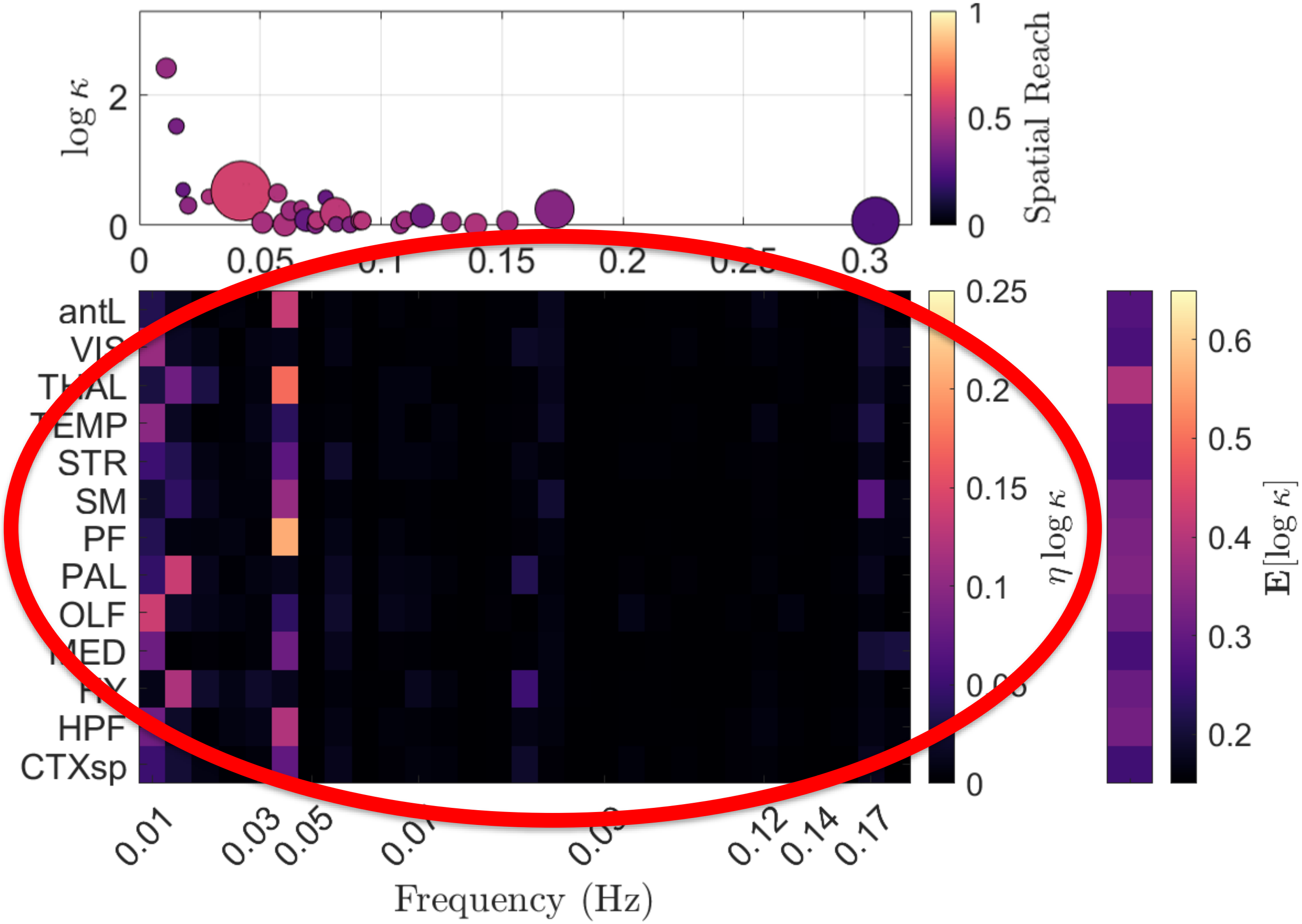
Modes with higher ellipticity are constrained to lower frequencies.



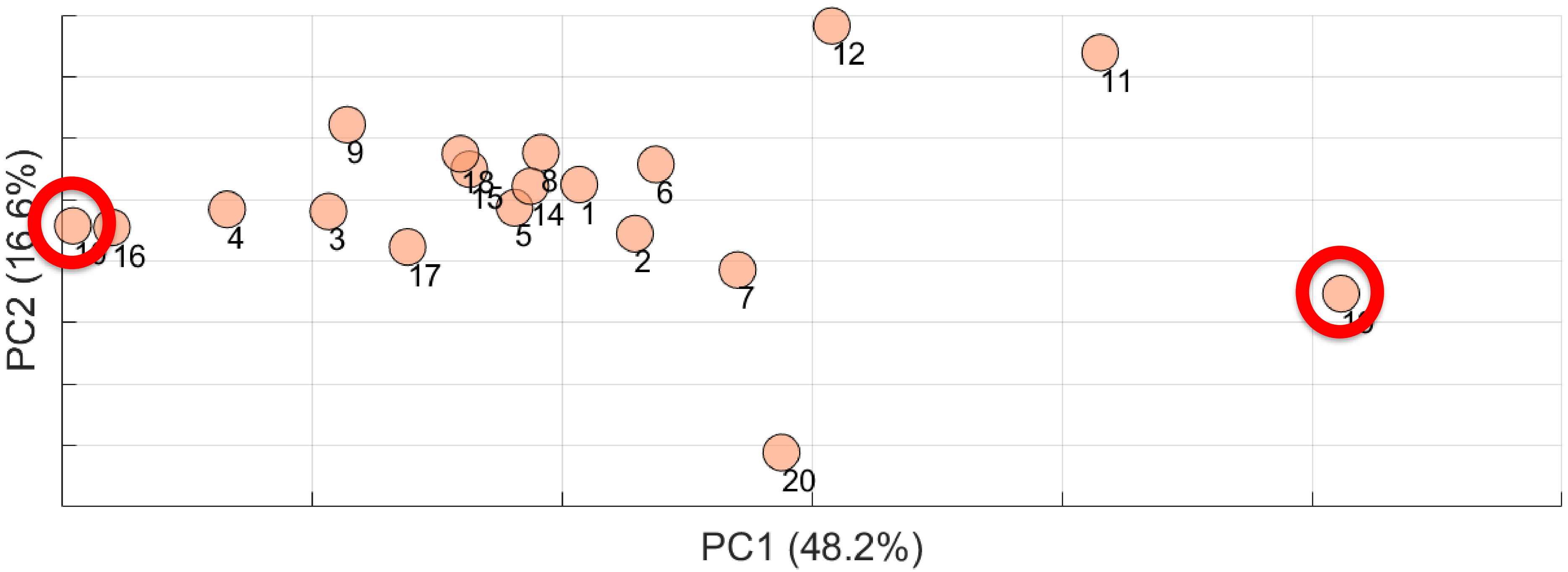
The spatial distribution of ellipticity differentiates subjects.

Invariant: regional expected ellipticity

Discriminant: regional ellipticity dispersion

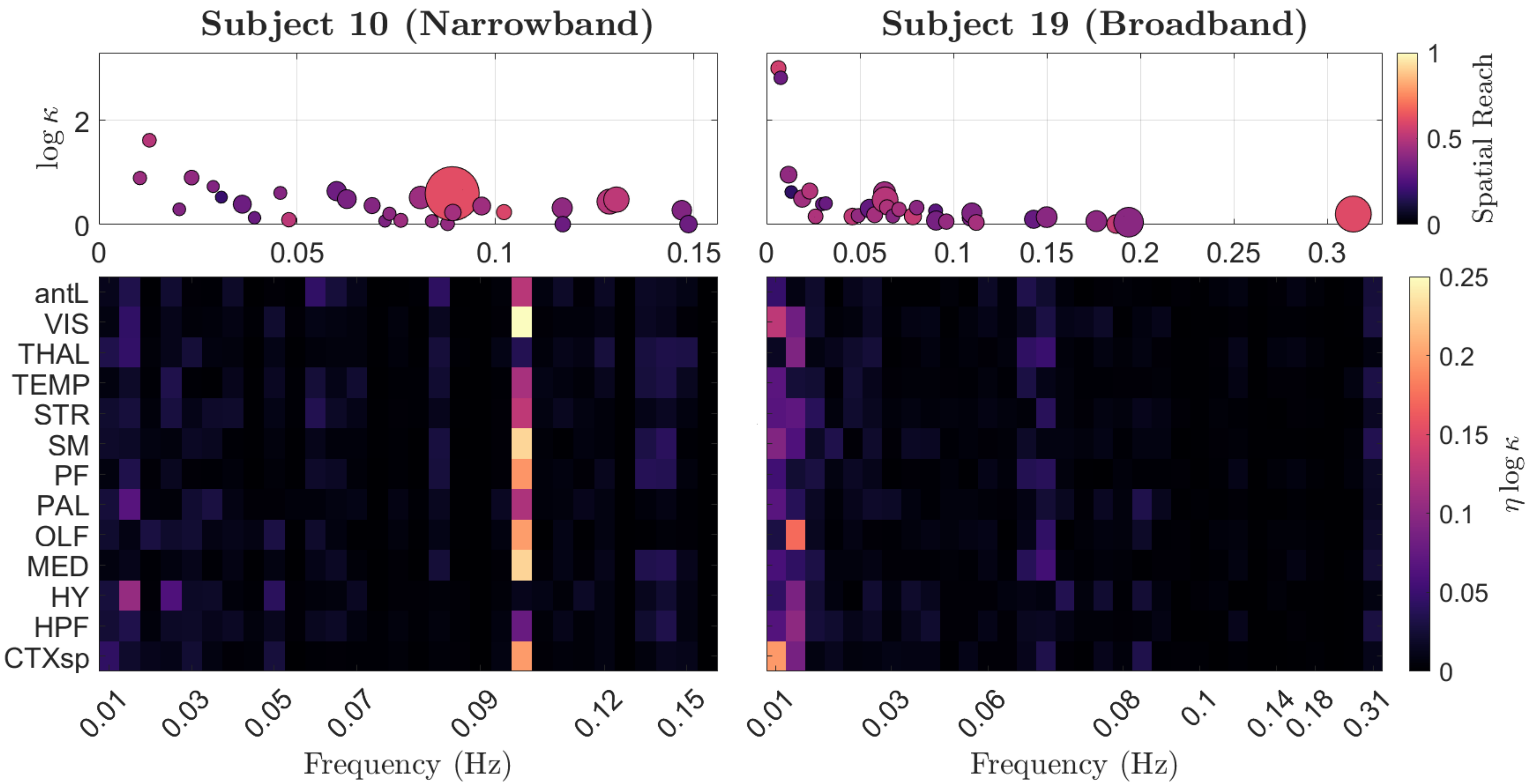


Principal component analysis isolates dynamical phenotypes.



The extremes of the first principal component identify two "dynamical phenotypes".

Dynamical phenotypes distribute differently ellipticity across brain regions.



Dynamical phenotypes might reflect functional segregation and integration.

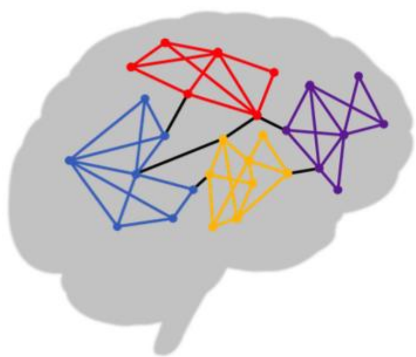
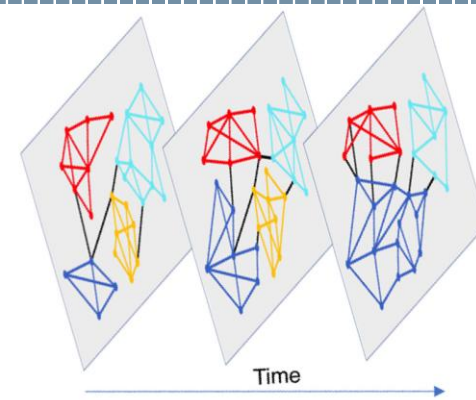
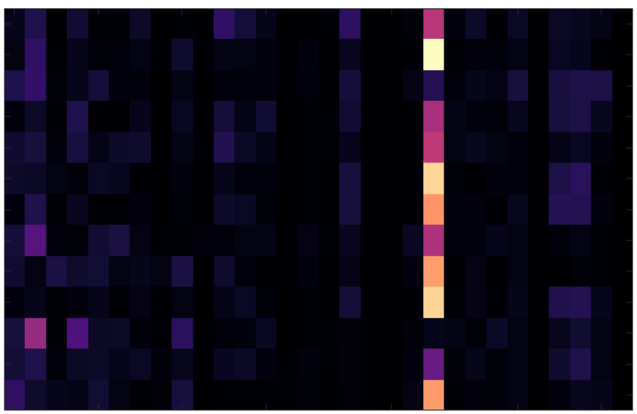
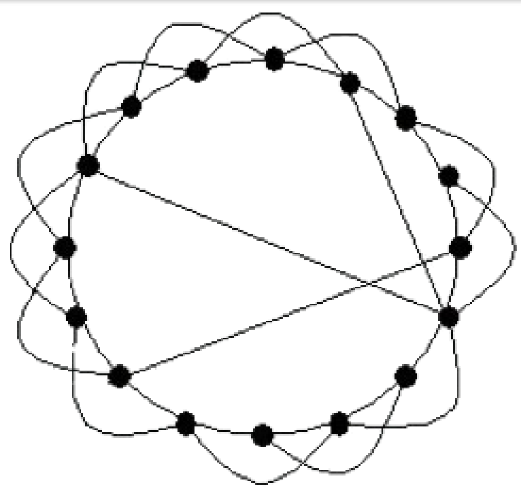


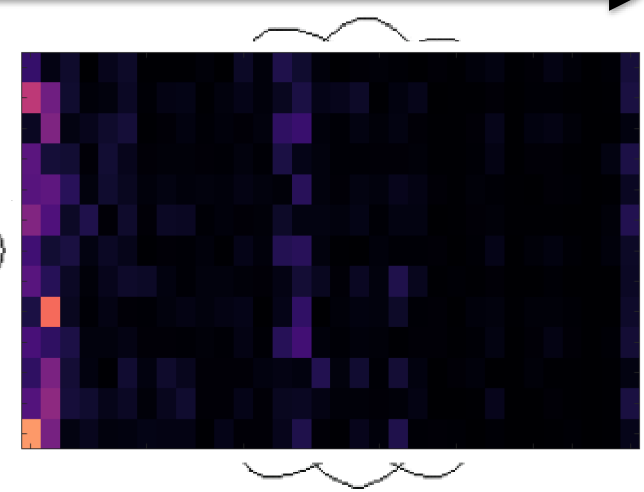
Figure: Tompson et al. (2018)
Personality Neuroscience



Segregation



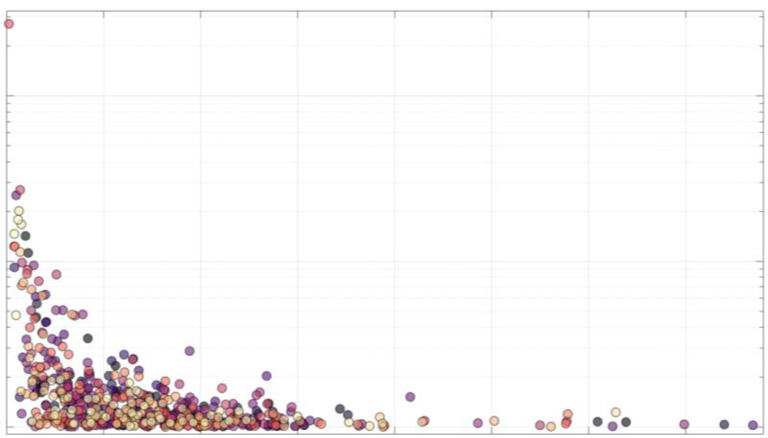




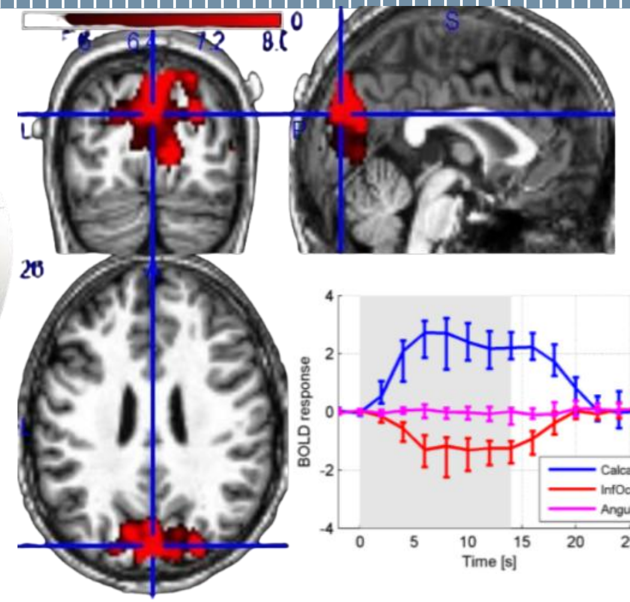
Integration

The frequency constraint could be caused by the nature of the BOLD signal.

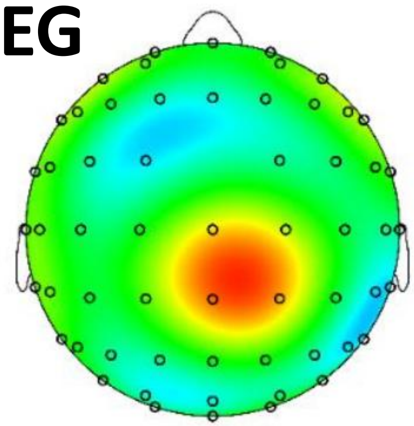
BOLD fMRI does not accurately capture high-frequency neural dynamics [4].

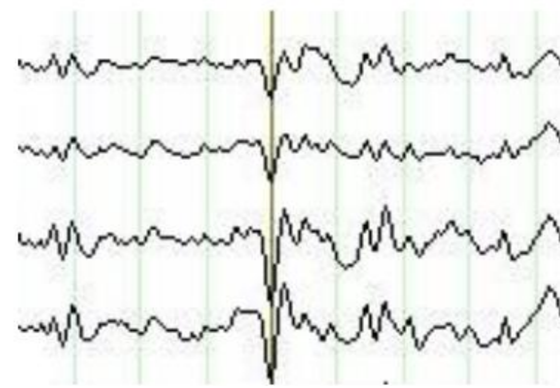


fMRI



EEG





Thank you for the attention.

References

1. Lynn, C. W., Cornblath, E. J., Papadopoulos, L., Bertolero, M. A., & Bassett, D. S. (2021). Broken detailed balance and entropy production in the human brain. *Proceedings of the National Academy of Sciences*
2. Nartallo-Kaluarachchi, R., Kringelbach, M., Deco, G., Lambiotte, R., & Goriely, A. (2026). Nonequilibrium physics of brain dynamics. *Physics Reports*
3. Benozzo, D., Baggio, G., Baron, G., Chiuso, A., Zampieri, S., & Bertoldo, A. (2024). Analyzing asymmetry in brain hierarchies with a linear state-space model of resting-state fMRI data. *Network Neuroscience*
4. Logothetis, N. K. (2003). The Underpinnings of the BOLD Functional Magnetic Resonance Imaging Signal. *The Journal of Neuroscience*

The state-interaction matrix was inferred under the sparse-DCM framework.

$$\dot{x}(t) = Ax(t) + w(t),$$
$$y(t) = h(x(t);\theta) + e(t)$$

Effective connectivity is estimated by inverting the DCM using fMRI data. In a Bayesian framework, this inversion cannot be computed in closed form.

The inversion process in sparse-DCM is framed as a maximum a posteriori estimation process.

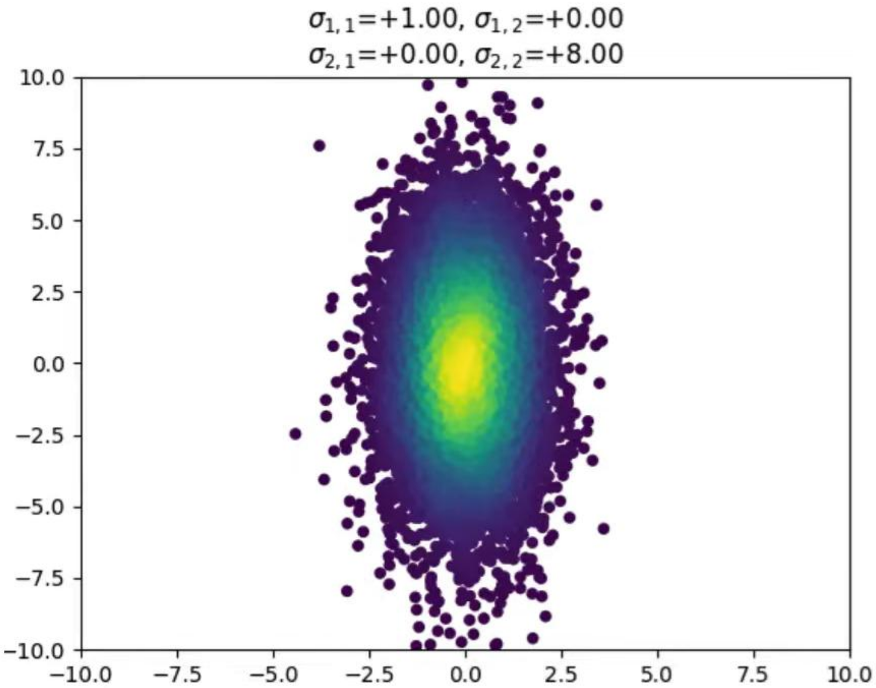
$$\hat{\theta} = \arg_{\theta} \max \ln p(Y|\theta) + \ln p_{\gamma}(\theta)$$

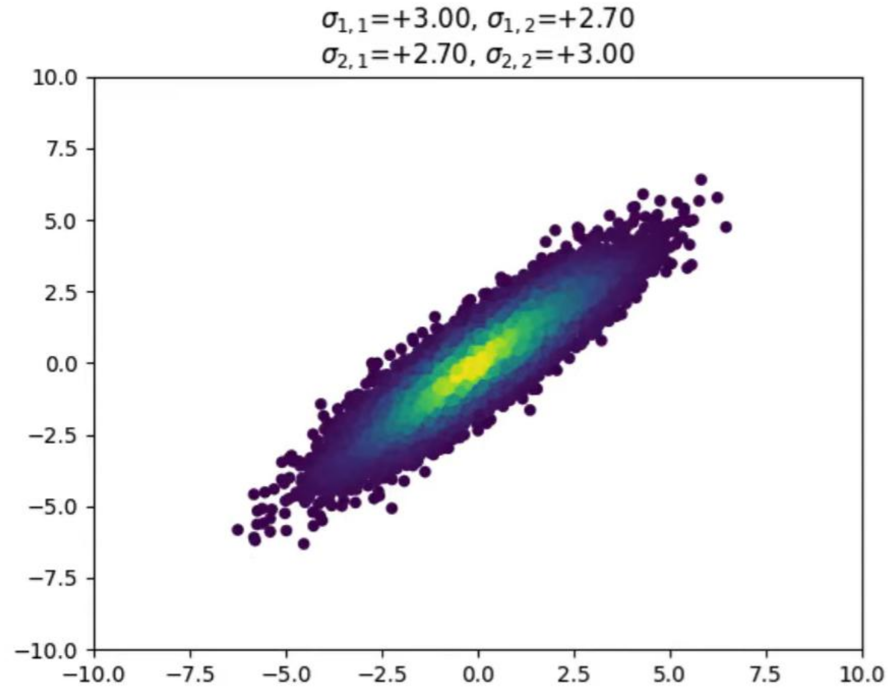
Because the hidden neural states cannot be observed directly, the inversion is performed using an expectation-maximization algorithm.

The covariance matrix (Σ) defines the shape of the potential landscape.

$$\Sigma = \begin{bmatrix} \boxed{\sigma_{1,1}} & \sigma_{1,2} \\ \sigma_{2,1} & \boxed{\sigma_{2,2}} \end{bmatrix}$$

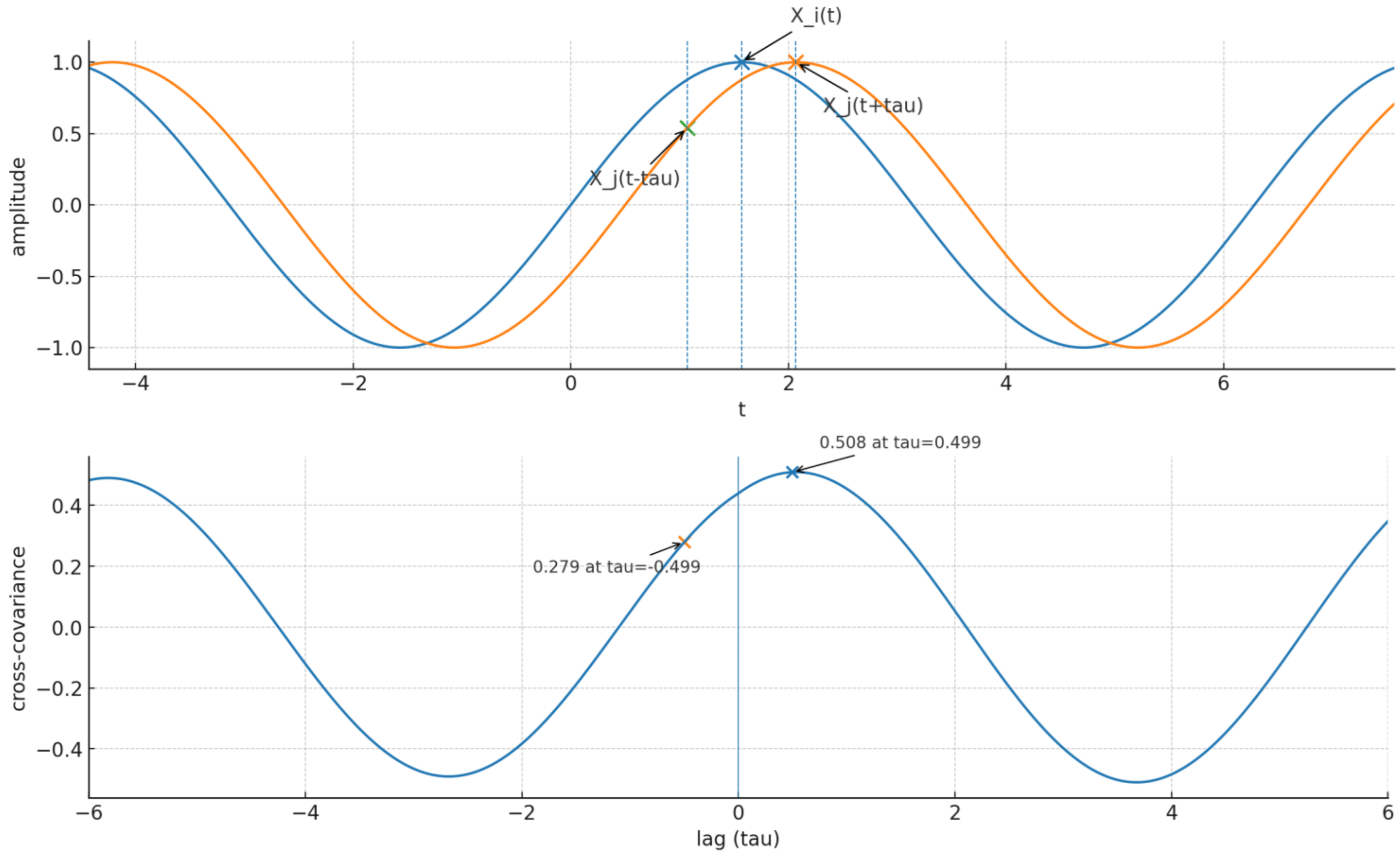
$$\Sigma = \begin{bmatrix} \sigma_{1,1} & \boxed{\sigma_{1,2}} \\ \boxed{\sigma_{2,1}} & \sigma_{2,2} \end{bmatrix}$$





The cross-covariance function measures lead-lag relationships between nodes.

$$\Sigma_{ij}(\tau) = \mathbb{E}\left[x_i(t+\tau)x_j(t)\right] = \mathbb{E}\left[x_i(t)x_j(t-\tau)\right] = \Sigma_{ji}(-\tau)$$



S enforces lead-lag relationships between nodes.

$$\frac{d\Sigma(\tau)}{d\tau} = A\Sigma(\tau)$$

$$\left.\frac{d\Sigma(\tau)}{d\tau}\right|_{\tau=0} = A\Sigma$$

$$\left.\frac{d\Sigma_{ij}(\tau)}{d\tau}\right|_{\tau=0} = S_{ij}$$

S breaks the symmetry of the cross-covariance function (defined by a null zero-lag derivative).

There can't be any transient growth in the whitened frame.

$$
\begin{aligned}
\frac{d|z(t)|}{dt} &= \frac{d}{dt}\left(z(t)^\top z(t)\right) \\
&= \dot{z}^\top z + z^\top \dot{z} \\
&= z^\top\left(\tilde{A} + \tilde{A}^\top\right)z \\
&= 2z^\top \widetilde{D} z < 0
\end{aligned}
$$

Because $\widetilde{D}$ is a negative definite matrix.

Ellipticity (κ) as a function of (conservative) rotational drive ($\widetilde{\omega}$).